

Interface Test Plan

FHIR R4 Expandable Platform · Generated 2026-08-05 from plan.json

This plan is the source of truth for automated verification. Run **python3 tools/run_interface_tests.py** to execute it and write **documents/test-results.json** + HTML. Update **tests/plan.json** as capabilities are added so the PDF and runner stay aligned.

1. Purpose

Automated verification for Phase 1–6: metadata, OAuth write protection, CRUD/search, Bundle transaction/batch, Bulk \$export, validation failures, Web UI chrome, and stakeholder documents.

2. Environments & base URLs

Key	URL
webui	http://127.0.0.1:8111
fhir	http://127.0.0.1:8110/eip/rest/fhir
token	http://127.0.0.1:8112/realms/fhir-demo/protocol/openid-connect/token
eip	http://127.0.0.1:8110/eip

- http://127.0.0.1:8111/
- http://127.0.0.1:8110/eip/rest/fhir/metadata
- http://127.0.0.1:8112/realms/fhir-demo

3. Test suites

Platform health

Core services are reachable before functional checks.

- **Web UI home loads** (ui → expect HTTP 200)
- **FHIR CapabilityStatement metadata** (http → expect HTTP 200)
- **Keycloak realm endpoint responds** (http → expect HTTP 200)

Authorization

- **Unauthenticated Patient create returns 401** (http → expect HTTP 401)
- **Fetch Keycloak client_credentials token** (oauth)

CRUD & search

- **Create Patient with Bearer** (http → expect body markers)
- **Read Patient pat-alice-001** (http → expect HTTP 200)
- **Search Patient family=Smith** (http → expect HTTP 200)
- **Create Observation** (http → expect body markers)
- **Search Observation by patient+code** (http → expect HTTP 200)

Bundle transaction/batch

- **Transaction Bundle persists entries** (http → expect HTTP 200)
- **Transaction Patient readable** (http → expect HTTP 200)
- **Batch Bundle returns per-entry statuses** (http → expect HTTP 200)
- **Batch Organization persisted** (http → expect HTTP 200)

Bulk \$export

- **\$export returns 202 + Content-Location** (http → expect HTTP 202)

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- **\$export-status eventually includes output** (wait)

Validation

- **Invalid Patient gender returns 400** (http → expect HTTP 400)

Web UI & documents

- **Routes API lists V2 routes** (http → expect HTTP 200)
- **Demo UI exposes FHIR ops controls** (ui → expect HTTP 200)
- **Capability Brief PDF served** (http → expect HTTP 200)
- **Route diagrams PDF served** (http → expect HTTP 200)

4. How to run

From the interface root (stack up):

```
python3 tools/run_interface_tests.py --wait
```

Watch mode (re-run on DESIGN/routes/samples/plan changes):

```
python3 tools/run_interface_tests.py --watch
```

After compose:

```
docker compose up -d --build && ./tools/post_up_tests.sh
```

Results appear in the Web UI Tests tab, and as documents/test-results.html (easy pass/fail list) plus documents/test-results.json.